STC89C52 MICROCONTROLLER

Register and Memory Documentation

1. STC89C52RC Memory

- The STC89C52RC is 80C51-compatible microcontroller.
- Provides 8 KB of on-chip Flash program memory
- 512 bytes of on-chip RAM.
- Its 8051-compatible data-memory organization includes the conventional 256-byte internal data-memory area.
- For the simulator, the conventional 8051 memory model should be kept clearly separated from any extended device-specific memory.

Memory / Area	Size / Addressing	Purpose
Program Flash	8 KB on-chip	Stores program/instruction code.
Internal data RAM	256-byte 8051 data space	Variables, register banks, bit-addressable area and general-purpose RAM.
Lower RAM	00H–7FH	Direct and indirect addressing.
Register banks	00H–1FH	Four banks, each containing R0–R7.
Bit-addressable RAM	20H–2FH	128 individually addressable bits.
Upper RAM	80H–FFH	128 bytes of upper RAM, accessed indirectly in the standard 8051 organization.
SFR space	80H–FFH	Control/status registers, accessed by direct addressing.
Stack	Uses internal RAM	Stores return addresses and temporary stack data; controlled by SP.

2. Internal RAM Organization

The lower 128 bytes of internal RAM are divided into three important regions.

- 32 bytes contain four register banks.
- 16 bytes are bit-addressable.
- 48 bytes are general-purpose RAM.

3. Register Banks and R0–R7

- The 8051 architecture contains four register banks.
- Each bank has eight 8-bit registers named R0 through R7.
- Only one bank is active at a time.
- The RS1 and RS0 bits of the Program Status Word (PSW) select the active bank.

RS1	RS0	Selected Bank	Address Range
0	0	Bank 0	00H–07H
0	1	Bank 1	08H–0FH
1	0	Bank 2	10H–17H
1	1	Bank 3	18H–1FH

4. Main CPU Registers

Register	Size	Address / Type	Function
ACC (A)	8-bit	E0H, SFR	Accumulator used extensively for arithmetic, logic and data movement.
B	8-bit	F0H, SFR	Used by MUL AB and DIV AB and can also be used as a general register.
PSW	8-bit	D0H, SFR	Contains CPU status flags and register-bank selection bits.
SP	8-bit	81H, SFR	Stack Pointer; points to the current top of the stack.
DPTR	16-bit	DPL=82H, DPH=83H	16-bit data pointer for addressing data/external memory.
PC	16-bit	CPU register	Program Counter; holds the address of the next instruction to fetch.
R0–R7	8-bit each	Internal RAM	Working registers in the currently selected register bank.

5. Stack and Stack Pointer

The stack is located in internal data RAM. The Stack Pointer (SP) is an 8-bit SFR at address 81H. It identifies the current stack position. During PUSH operations the stack grows upward in internal RAM. CALL/ACALL instructions and interrupts also use the stack to preserve return addresses. POP and RET/RETI operations retrieve stack data as appropriate.

- After reset, SP has the standard 8051 reset value of 07H.
- A PUSH operation increments SP before storing the byte.
- A POP operation reads the byte at SP and then decrements SP.
- CALL/ACALL and interrupts push return information onto the stack.
- The simulator should detect stack growth beyond the available internal RAM region.

6. Special Function Registers (SFRs)

- SFRs are control and status registers located in the 80H–FFH address range.
- They control ports, timers, serial communication, interrupts and CPU functions.
- SFRs are accessed using direct addressing.

SFR	Address	Main Purpose
P0	80H	Port 0 latch / I/O control
SP	81H	Stack Pointer
DPL	82H	Low byte of DPTR
DPH	83H	High byte of DPTR
PCON	87H	Power-control functions
TCON	88H	Timer 0/1 control and external interrupt control
TMOD	89H	Timer 0/1 mode selection
TL0 / TL1	8AH / 8BH	Low timer/counter bytes
TH0 / TH1	8CH / 8DH	High timer/counter bytes
P1	90H	Port 1 latch / I/O control
SCON	98H	Serial port control
SBUF	99H	Serial data buffer
P2	A0H	Port 2 latch / I/O control
IE	A8H	Interrupt enable control
P3	B0H	Port 3 latch / I/O control
IP	B8H	Interrupt priority control
PSW	D0H	Program status word
ACC	E0H	Accumulator
B	F0H	B register

7. Bit-Addressable Memory and SFRs

The internal RAM area 20H–2FH contains 128 individually addressable bits. This is useful for Boolean variables and bit-oriented instructions. In addition, selected SFRs are bit-addressable. A bit-addressable SFR is one whose SFR address is divisible by 8 in the standard 8051 map.

- Bit-addressable RAM: byte addresses 20H–2FH.
- This area provides 128 individual bits.
- Important bit-addressable SFR examples include P0, TCON, P1, SCON, P2, IE, P3, IP and PSW.
- The simulator should support both byte-level and bit-level access where the instruction set requires it.

8. DPTR and Data Addressing

- DPTR is a 16-bit data pointer made from DPH and DPL.
- DPH contains the high byte and DPL contains the low byte.
- The STC89C52RC also supports a second data pointer through its device-specific DPTR switching mechanism.
- For a first version of the educational simulator, DPTR0 can be modeled as the primary 16-bit pointer, with DPTR1 added if the team implements the enhanced feature.